

Day02

day02.pdf | GPIO PWM

1. GPIO

GPIO General Purpose Input/Output

	→		LED
	→		ADC

500ms + 40us -
DATA

GND GPIO 1 - GND
 →

2.

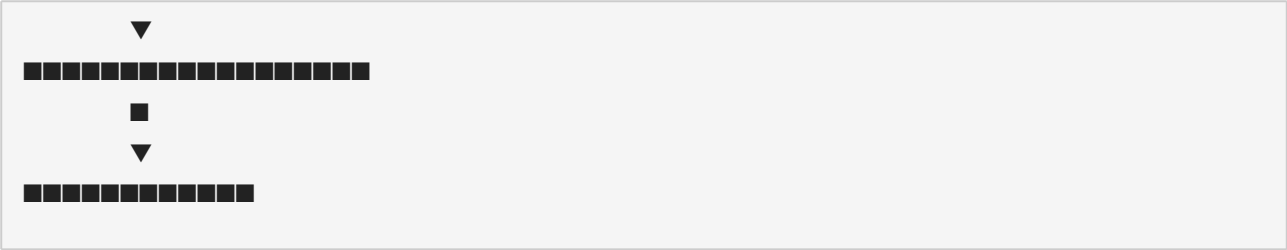
CPU CPU

CPU / - CPU
 - CPU
51

3. 51

51 5

|--|--|--|



0x00000000 - 0x00000000 CPU 0x00000000 - 0x00000000

6. 51 定时器

51 定时器 2 个 - Timer0 0 - Timer1 1

16 位寄存器 16 位寄存器 - 65535 unsigned short 0~65535
- TL 8 位 TH 8 位

0x00000000 - 0x00000000

7. 中断

0x00000000 - 0x00000000

位	寄存器	范围	位
16 位寄存器	16 位 TL + TH	0 ~ 65535	0x00000000 - 0x00000000
8 位寄存器	8 位 TL 寄存器	0 ~ 255	0x00000000 TH 0x00000000 TL 0x00000000

8 位寄存器 UART 0x00000000

8. 串口

1ms 12MHz

0x00000000

```

12MHz
512 = 12MHz / 12 = 1MHz
1 = 1 / 1MHz = 1us

1ms = 1000us
1000 = 1000us / 1us = 1000
65535 = 65535 - 1000 = 64535

```

1. 1

1. TL TH
2. TL = 64535 & 0xFF
3. TH = 64535 >> 8
- 4.
5. +1
6. 64535 + 1 → → CPU

1

9.8

1

```

(255→0)
TH → TL
( ) ( )
0→255

```

- TL +1
- TH
- TL 255 + 1 → TH TL
-

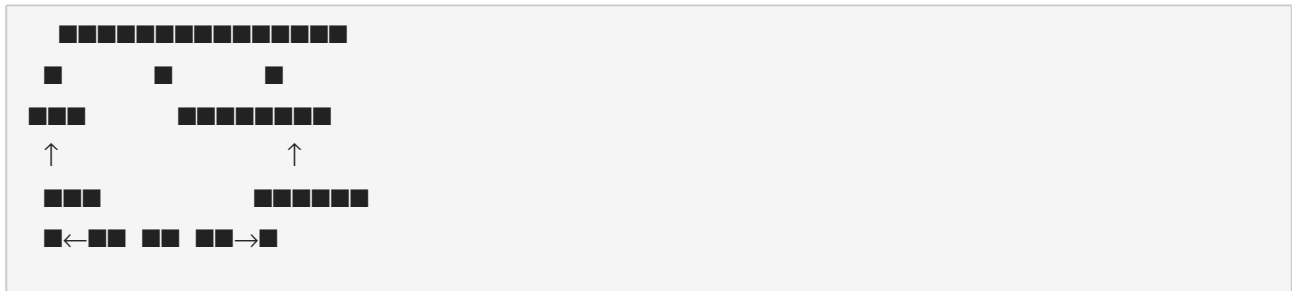
16 - 16 TL TH - 8

UART

10. PWM

PWM Pulse Width Modulation

PWM



11. PWM

PWM



- 50%
- → LED

LCD LED

200Hz PWM 50% 12MHz

PWM 200Hz

$PWM = 1 / 200Hz = 0.005s = 5ms$

50% = $5ms / 2 = 2.5ms$

$12MHz / 12 = 1MHz \rightarrow 1\mu s$

$2.5ms / 1\mu s = 2500$

$65535 - 2500 = 63035$

12.

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